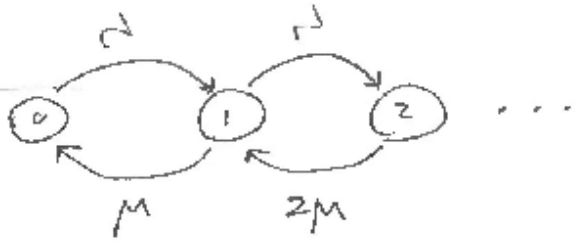
E4.05 - The Answers 2024

Q1
a)



[2]

Birth coeff:

$P(N_{t+\Delta t} = i+1 \mid N_t = i) = \lambda \Delta t$ for all i

$\boxed{\lambda_i = \lambda}$

Death coeff:

i channel busy at time t

$P(1 \text{ channel} \rightarrow \text{idle}) = \binom{i}{1} (\mu \Delta t)(1-\mu \Delta t)^{i-1}$

$= i\mu \Delta t + o(\Delta t)$

$\boxed{\mu_i = i\mu}$

[2]

Equilibrium

$\pi_i = \left(\frac{\lambda_{i-1}}{\mu_i}\right) \pi_{i-1}$ $\qquad \rho = \frac{\lambda}{\mu}$

$= \left(\frac{\lambda}{i\mu}\right) \pi_{i-1} = \left(\frac{\rho}{i}\right) \pi_{i-1}$

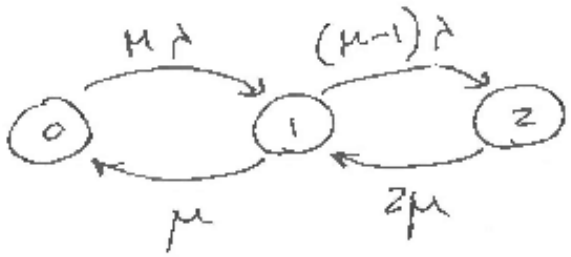
$= \left(\frac{\rho^i}{i!}\right) \pi_0 \quad i = 1, 2, \cdots$

$\pi_0 = \frac{1}{S} = e^{-\rho}$

$\pi_i = \frac{\rho^i}{i!} e^{-\rho} \quad i = 0, 1, 2, \cdots$

[3]

Q1 b)



[2]

Birth coeff

Number of idle sources in state i is (M-i)

$$\lambda_i = (M-i)\lambda = \left(1-\frac{i}{M}\right)\lambda_0 \qquad \lambda_0 = M\lambda$$

Death coeff

$$\mu_i = i\mu.$$

Equilibrium [2]

$$\pi_i = \left(\frac{\lambda_{i-1}}{\mu_i}\right)\pi_{i-1} = \left(\frac{(M-(i-1))\lambda}{i\mu}\right)\pi_{i-1}$$

$$= \binom{M}{i}\alpha^i \pi_0 \qquad i = 1, 2, \ldots M$$

$$\pi_0 = \left[\sum_{j=0}^{M}\binom{M}{j}\alpha^j\right]^{-1}$$

$$p = \frac{\alpha}{1+\alpha} \longleftrightarrow \alpha = \frac{p}{1-p}$$

$$\pi_i = \binom{M}{i} p^i (1-p)^{M-i}$$

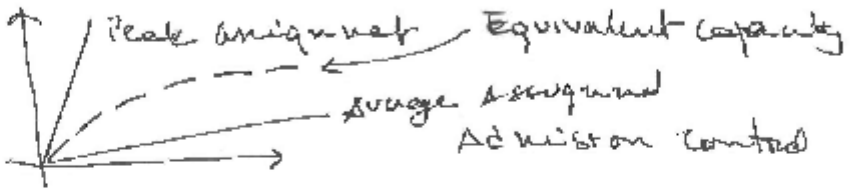
[3]

Q1c)

i) - Residual time = Residual service time seen by a virtual arrival at time t.

- $E[R_t] = \frac{1}{2} \lambda E[S^2]$ and its relevance when deriving $E[w]$ and priority queuing systems [2]

ii)



- discuss trade-off and multiplexing gains [2]

iii) $P(X=x) = \frac{(\lambda t)^x e^{-\lambda t}}{x!}$

$P(x=0) = e^{-\lambda t} = 1 - \lambda t + \frac{\lambda t^2}{2!}$

$= 1 - \lambda t + o(t)$

$P(x \geq 1) = 1 - P(x=0)$

$= 1 - 1 + \lambda t - o(t)$

$= \lambda t - o(t)$

$\approx \lambda t \qquad t = \Delta t$ [2]

Q2a)



[3]

$$\pi_i = \left(\frac{A}{i}\right)\pi_{i-1} \qquad i \le K \qquad \rho = \frac{A}{K}$$

$$\pi_i = \left(\frac{A}{K\mu}\right)\pi_{i-1} = \rho\,\pi_{i-1} \quad i \ge K \qquad A = \frac{\lambda}{\mu}$$

$$\pi_i = \left(\frac{A^i}{i!}\right)\pi_0 \qquad i \le K$$

$$\left(\frac{A^K}{K!}\right)\rho^{i-K}\pi_0 \qquad i \ge K \qquad , \pi_K = \left(\frac{A^K}{K!}\right)\pi_0$$

$$i \ge K \qquad i = K+j \qquad \pi_{K+j} = \left(\frac{A^K}{K!}\right)\rho^j\pi_0$$

$$= \rho^j \pi_K$$

$$S = \sum_{i=0}^{K}\left(\frac{A^i}{i!}\right) + \left(\frac{A^K}{K!}\right)\left(\frac{\rho}{1-\rho}\right)$$

$$\pi_0 = \frac{1}{A^K/K!}\left[\frac{(1-\rho)E_K(A)}{(1-\rho) + \rho E_K(A)}\right]$$

[5]

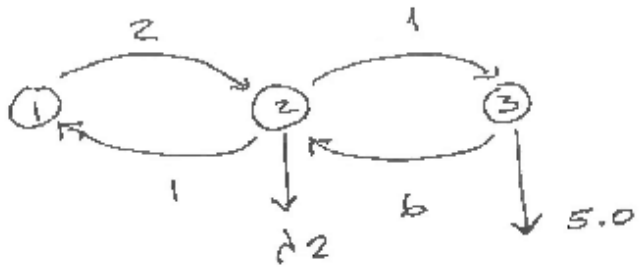
$$P(Q_t = i \mid n_K \ge K) = \frac{P(n_t = K+i)}{\sum_{j=0}^{\infty} P(n_t = K+j)}$$

$$= \frac{\pi_K \rho^i}{\sum_{j=0}^{\infty} \pi_K \rho^j}$$

$$P(Q_t = i \mid \text{delay}) = (1-\rho)\rho^i \qquad i = 0, 1, 2 \ldots$$

[3]

Q2(5)



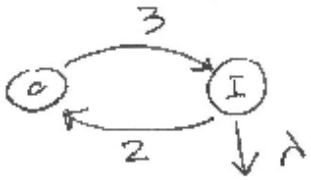
$$Q = \begin{bmatrix} -2 & 2 & 0 \\ 1 & -2 & 1 \\ 0 & 6 & -6 \end{bmatrix}$$

$$Q^T \pi = 0, \quad \pi e = 1 \longrightarrow \begin{bmatrix} -2 & 1 & 0 \\ 2 & -2 & 6 \\ 0 & 1 & -6 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

$x_1 = \frac{3}{10} \quad x_2 = \frac{6}{10} \quad x_3 = \frac{1}{10}$

mean rate over log run $= 3.2 = \frac{6}{10}\lambda_2 + \frac{1}{10}5.0$

$2.7 = \frac{6}{10}\lambda_2 \longrightarrow \lambda_2 = \frac{27}{6} \Rightarrow \boxed{\lambda_2 = 4.5}$ [4]



$\pi_{OFF} = \frac{2}{3+2} \quad \pi_{ON} = \frac{3}{3+2}$

mean rate over log run $= 3.2 = \frac{3}{5}\lambda$

$\lambda = \frac{16}{3} \rightarrow \boxed{\lambda = 5.\overline{3}}$ [3]

Discussion
- model degrees of freedom
- complexity
- approximation [2]

Q3

i) Total arrival rate = 225 + 75 = 300 messages/s

Mean length of packet = $0.75 \times 160 + 0.25 \times 320$

$= 1.25 \times 160$

Mean squared length = $0.75 (160)^2 + 0.25 (320)^2$

$= 0.75 (160)^2 + 0.25 \times 4 (160)^2$

$= 1.75 (160)^2$

$$E(S) = \frac{1.25 \times 160}{64} = \frac{25}{8} = 3.125 \text{ s}$$

$$E(S^2) = 1.75 \frac{(160)^2}{(64)^2} = 10.9375 \text{ s}^2$$

Using

$$E(W) = \frac{\lambda E(S^2)}{2(1-\rho)} = 26.4 \text{ ms}$$

$$E(T) = E(W) + E(S) = 29.5 \text{ ms}$$ [6]

Q3

ii)

$$\rho_1 = \lambda_1 E(S_1) = 225 \times \frac{160}{64} = 0.5625$$

$$\rho_2 = \lambda_2 E(S_2) = 75 \times \frac{320}{64} = 0.375$$

$$R = \frac{1}{2}\lambda E(S^2) = 1.65 \text{ ms}$$

$$E(W_1) = \frac{E(R)}{1-\rho_1} = \frac{1.65}{1-0.5625} = 3.77 \text{ ms}$$

$$E(T_1) = E(W_1) + E(S_1) = 6.27 \text{ ms} \quad [3]$$

$$E(W_2) = \left[\frac{E(W_1)}{1-\rho_1-\rho_2}\right] = \frac{3.77}{1-0.9375}$$

$$= 60.3 \text{ ms}$$

$$E(T_2) = E(W_2) + E(S_2) = 65.3 \text{ ms} \quad [2]$$

$$E(T) = \frac{3}{4} E(T_1) + \frac{1}{4} E(T_2) = 21.0 \text{ ms} \quad [3]$$

iii)

Q3 ii)

class 1 $\rho_1 = \lambda_1 E(s_1) = 0.5625$

$E(w_1) = 3.77$ ms

class 2 $\rho_2 = \lambda_2 E(s_2) = 0.375$

$E(w_2) = 60.3$ ms

$$\sum_i \rho_i E(w_i) = 2.12 + 22.62 = 24.74 \text{ ms}$$

Here $\rho = 0.9375$

$E(R) = 1.65$ ms

$$\frac{\rho}{1-\rho} E(R) = 15 \times 1.65 = 24.74 \text{ ms}$$

[6]

Q4e)

If i sources on $\Rightarrow \alpha i$

the system has equivalent capacity αC



$$\frac{\Delta x}{\Delta t} = (i - C)\alpha$$

$$\Delta x = (i - C)\alpha \Delta t \quad \left(\dot{x} - \alpha C = \text{"rate of filling"}\right) \qquad [2]$$

ii) the condition for the buffer to be depleting is

$$i - C < 0 \quad \Rightarrow \quad i < C \qquad [2]$$

iii) $F_i(t+\Delta t, x)$ = probability that buffer occupancy is less than or equal to x while i sources on (at time $t+\Delta t$)

$$F_i(t+\Delta t, x) = [N-(i-1)]\lambda \Delta t F_{i-1}(t,x)$$
$$+ (i+1)\alpha \Delta t F_{i+1}(t,x)$$
$$+ \{1 - [(N-i)\lambda + i\alpha]\Delta t\} F_i[t, x - \underbrace{(i-C)\alpha\Delta t}_{\Delta x}]$$
$$+ o(\Delta t)$$

$$\Delta x = (i-C)\alpha\Delta t$$
$$h_1 = [N-(i-1)]\lambda$$
$$h_2 = (i+1)\alpha$$
$$h_3 = -[(N-i)\lambda + i\alpha]$$

$$F_i(t+\Delta t, x) - F_i(t,x) = h_1 \Delta t F_{i-1}(t,x) + h_2 \Delta t F_{i+1}(t,x) + h_3 \Delta t F_i(t, x-\Delta x) + F_i(t, x-\Delta x) - F_i(t,x)$$

$$\frac{F_i(t+\Delta t, x) - F_i(t,x)}{\Delta t} = h_1 F_{i-1}(t,x) + h_2 F_{i+1}(t,x) + h_3 F_i(t, x-\Delta x) + \frac{F_i(t, x-\Delta x) - F_i(t,x)}{\Delta x} \frac{\Delta x}{\Delta t}$$

$$\frac{\partial F_i(t,x)}{\partial t} = h_1 F_{i-1}(t,x) + h_2 F_{i+1}(t,x) + h_3 F_i(t, x-\Delta x) - \left(\frac{F_i(t,x) - F_i(t, x-\Delta x)}{\Delta x} \right) (i-c)\alpha$$

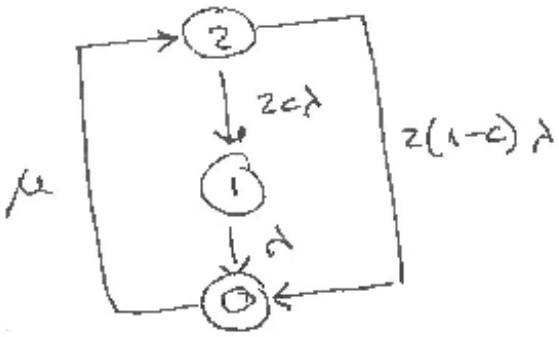
In $\Delta x \to 0$

$$\frac{\partial F_i(t,x)}{\partial t} = h_1 F_{i-1}(t,x) + h_2 F_{i+1}(t,x) + h_3 F_i(t,x) - (i-c)\alpha \frac{\partial F_i(t,x)}{\partial x}$$ [5]

iv) $$\frac{\partial F_i(t,x)}{\partial t} = 0$$ [2]

Q4(b) i) The transition $N(1-c)\lambda$ represents a simultaneous failure of two (2) computer [2]

ii)



$\mu = 6$
$\lambda = 2$
$c = 0.999$ [2]

$$Q = \begin{bmatrix} -2\lambda & 2c\lambda & 2(1-c)\lambda \\ 0 & -\lambda & \lambda \\ \mu & c & -\mu \end{bmatrix} = \begin{bmatrix} -4 & 3.996 & 0.004 \\ 0 & -2 & 2 \\ 6 & c & -6 \end{bmatrix}$$

$$Q^T = \begin{bmatrix} -4 & 0 & 6 \\ 3.996 & -2 & 0 \\ 0.004 & 2 & -6 \end{bmatrix} \begin{bmatrix} \pi_1 \\ \pi_2 \\ \pi_3 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

$R_{steady\ state} = R_1\pi_1 + R_2\pi_2 + R_3\pi_3$ [3]

iii) $R = [1\ 1\ 0]$

1 = operational state
0 = failure state [2]